

Logistics

- We meet Fridays, 13-15h. Please try to attend in person!
- I will try to remember to record the classes and put them on YouTube
- To email the group: CLUBclub.virologie@charite.de
- Chat using Slack (ask me for an invite) in our `#club-club` channel
- These slides are on GitHub, at <https://github.com/VirologyCharite/club-club>
- Cluster help: <https://git.bihealth.org/charite-sc-public/sc-wiki/-/wikis/home>
- Getting a terminal (when on the VPN): <https://s-sc-hub.charite.de/>

You need to be on the Charité VPN if you're not on the wired Charité network.

Some motivational words

- Using the command line effectively is a big subject, even before we get to bioinformatics
- You have to do things. Don't be passive
- Help each other!
- We are not bioinformatics experts - we're still learning too
- There is a ton of help around online (YouTube, AI)

Course outline

There is an assumption that you're all interested in working with genetic sequences.

Geneious is a great piece of software. But sometimes you'll have more data or need more processing power than your laptop/desktop can supply, or you'll need to do something that's not possible in Geneious, or you simply won't have Geneious at all.

- Command-line basics (today)
- NGS data
- Getting sequences from NCBI (and maybe other databases)
- Mapping reads to references (nucleotide level)
- Mapping reads to references (protein level)
- De novo assembly
- Multiple sequence alignments
- Calling consensus
- Phylogenetic trees
- Using Claude to write you bioinformatics tools

Most of these conceptually simple tasks are actually very complicated!

Your suggestions?

- Using `git` and GitHub

Today

Will be a bit boring. I will cover some basics and get you to type a few one-time commands to get you fully set up.

You won't actually do anything much until next week.

High-performance compute system (HPCS)

- aka a "cluster"
- An often large set of conneced high-powered computers
- Running some version of UNIX
- With a job scheduler (SLURM)
- You can't really run graphical programs, you just have the terminal

What is the "terminal"?

- What is "the shell", "Terminal", the "command line", "bash"?
- All the commands below are to be entered in the Terminal on the cluster!

What is (was!) a terminal, anyway?

Top right: a Teletype **ASR 33**. I used one occasionally as an undergrad at Sydney University, in 1982!

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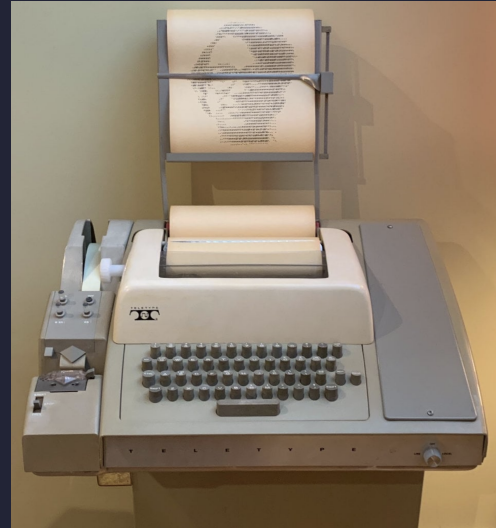
See

<https://www.harley.com/unix-book/book/chapters/03.html#C>

Bottom right: A much more modern cathode-ray tube (CRT) terminal, the famous **VT100** (introduced 1978).

See

https://en.wikipedia.org/wiki/Cathode_ray_tube#History



Some first Terminal commands

- `ls` - list files in a directory (use `ls -l` for a long listing)
- `cd` - change directory
- `pwd` - print working directory
- `man xxx` - read the manual entry for a command (here `xxx`)
- `cat` - show a file's content (on your terminal)
- `less` - show a file's content page by page. Use space to advance a page, 'q' to quit
- `cp` - copy a file
- `mv` - move (i.e., rename) a file
- `mkdir` - make a directory
- `rmdir` - remove a directory (must be empty)
- `control-d` - terminate input
- `control-c` - interrupt a program

Other useful commands

- `grep` - show lines that match a pattern
- `tr` - change characters into other characters
- `cut` - select fields or character ranges from lines

There are many hundreds of other UNIX commands!

Some command line syntax

- `/` is used to separate folder names when you give a path to a file
- A `/` by itself refers to the very top of the file hierarchy
- You are always "in" some folder (aka a "directory"). `pwd` will show you which
- File names can be relative to your current directory, or absolute (starting with `/`)
- `~` (tilde) refers to your home directory
- `.` refers to the current directory
- And `..` to the directory one up from where you are (if you're not in `/`)
- `*` wildcards can be used for filename matching
- `|` pipes for connecting program output to input
- `<` and `>` for input and output redirection
- The shell is a programming environment, with
 - Variables
 - Simple loops

Using the command-line efficiently

- Use `TAB` to complete file names
- Use up- and down- arrow to move in your command history
- Use `ESC .` to insert the last word from the last command

Learn to use a text editor

E.g., `nano` or `vim` or `SublimeText`, etc. There are many.

Note: Microsoft word is not a text editor!

Install Pixi

We will use Pixi (<https://pixi.prefix.dev/latest/>) for installing software packages. It runs on top of `conda` in case you're wondering.

Type this (exactly!) to install it in your cluster account.

```
$ cd  
$ curl -fsSL https://pixi.sh/install.sh | sh
```

Setting up Pixi

Edit your `~/.bashrc` file with `nano`:

```
$ nano ~/.bashrc
```

and add this line:

```
export PIXI_CACHE_DIR=/tmp/pixi-cache-$USER
```

Then tell your shell to re-read that file:

```
$ source ~/.bashrc
```

If everything has worked, `which pixi` should result in the following:

```
$ which pixi
~/.pixi/bin/pixi
```

Make a workspace and tell Pixi it can use bioconda

First (one-time setup), make a Pixi workspace called "workspace" and change directory into it:

```
$ pixi init workspace  
$ cd workspace
```

Edit `pixi.toml` and add `"bioconda"` to the channel list:

```
$ nano pixi.toml  
  
# You need to make this change:  
channels = ["conda-forge", "bioconda"]
```

Use Pixi to install some programs

Add `bowtie2`, for example:

```
$ pixi add bowtie2
$ pixi shell
$ which bowtie2
~/workspace/.pixi/envs/default/bin/bowtie2
```

Add our `dark-matter` package:

```
$ pixi add --pypi dark-matter
```

Shared storage

Our group has shared work and scratch space.

```
# 1TB, with snapshots and external backup  
/sc-projects/sc-proj-cc11-civclub
```

```
# 5TB, no snapshots, no backups  
/sc-scratch/sc-scratch-cc11-civclub
```

Using the shared storage

Make yourself a directory in the work area and change directory into it.

```
$ mkdir /sc-projects/sc-proj-cc11-civclub/terry
$ cd /sc-projects/sc-proj-cc11-civclub/terry
```

Assuming you have done the above, you might like to make a symbolic link for easier access:

```
$ cd # Change to your home directory
$ ln -s /sc-projects/sc-proj-cc11-civclub/terry work
$ cd work
$ pwd
/home/jonestc/work

# But where are we, really?
$ /bin/pwd
/sc-projects/sc-proj-cc11-civclub/terry
```

SMB access to the shared storage

If you need to access the file shares outside of the HPC environment in the Charité network, you can connect them as "Windows Network Drives" (SMB). The paths are:

```
\\sc-data.sc-store.charite.de\sc-project-cc11-civclub  
\\sc-data.sc-store.charite.de\sc-scratch-cc11-civclub
```